



Financial Institutions & Markets- ENT 309

PPT #10, IITD, Summer, 2026

Agenda

1

INSURANCE

- Insurance Concepts
- Insurance Market in India
- Cash Flows, Investments & Stability

PENSION

Insurance: Concepts



Insurance: Basics

- What is Insurance?
 - A risk transfer mechanism
 - "Financial" risk which can be measured and priced.
 - Insurance company takes over the risk in return for a 'consideration/fee/charges' called Insurance Premium.
 - Future "financial" impact borne by the "buyer" of risk/ insurance co.
- Insured/ Policy Holder: One whose life has been insured or who has bought the policy.
- Insurer: One who charges premium and buys the risk. Ex. Insurance Co.
- Agent "of the company"- Authorized by the Insurance Co. to act on its behalf and promote its products.
- Broker "of the buyer"- Authorized by the buyer to act on her behalf to get the best deal.



Insurance: Basics

- Sum Assured: Used only in Life Insurance
 - Is **the guaranteed payout** in life insurance (if the person expires)
- Sum Insured: Used in General Insurance
 - The **maximum amount** applicable to an insurance policy
 - Car insured till 4 lacs. Payout done as per damage assessed.
- Third Party Insurance: Used in car insurance
 - Protects the driver/owner from any liabilities arising from hurt/loss to anyone else.
 - Insurance Policy Holder: 1st Party, Insurance Co.: 2nd Party, Anyone else who suffers a loss due to 1st party is 3rd Party



Insurance: Basics

- Life Insurance: Pays a guaranteed amount if the insured expires.
 - Risk: Two types.
 - Dying early. Living too long.
 - Name a Life Insurance Co. in India
- Health Insurance: Pays specifies sums for medical expenses or treatments
 - Risk: Health issues causing huge financial loss.
 - Name a Health Insurance Co. in India
- General Insurance: Includes many types of insurance products
 - Car, House, Fire, Property, Marine, Travel, Gadget etc.
 - Includes Health Insurance.
 - Name a General Insurance Co. in India



Reinsurance

- Insured (Policyholder) transfers the risk to an insurance company.
- The insurance company – Life or General- can keep a part of the risk with it and can transfer the remaining risk to (buy insurance from) a reinsurer.
- Insurance co. pays a premium to Reinsurance co. in proportion to the risk transferred to it. Keeps the rest of the premium with itself.
- Could be done at an individual policy level (for large insurance policies) or at a portfolio level.
- Examples of Reinsurers are: Munich Re, Swiss Re, Allianz Re, GIC Re (General Insurance Company of India) etc.
- Reinsurance helps an insurance company insulate itself from big losses due to sudden increase in claims due to one-off events. Ex: COVID-19
- Most reinsurers are foreign companies. So, reinsurance results in substantial outflow of FX from India. Govt encouraging Indian Re companies.



Insurance: Basics

- Nominee: The person designated by a policyholder to receive the death benefit payout in the event of their demise.
- Claim: Intimation to the insurance company that an event causing a loss has occurred.
- Underwriting: Select applicants. Measure and price risk. Accept applicants. Reject other applicants.
- Policy Surrender: Return the insurance policy to the insurance co. before the maturity date. Get pre-decided amount from the company
 - Similar to _____ in bonds?
 - Puttable bonds.



Insurance: Basics

- Can you buy a Life insurance on the life of an unknown person X who lives in a house 5 houses away from your house?
- You can pay the premium and tell the insurance co. that if anything happens to X, they should pay you the Sum Assured to you.
- You **cannot** because you do not have “insurable interest” in X.
- A **person has an insurable interest** in something when loss-of or damage to that thing would cause the person to suffer a financial loss.



Insurance: Basics

A RISK is INSURABLE IF

- Policyholder or nominee have financial interest in the risk being insured,
- Policyholder has a personal interest in the claim event not happening to distinguish between insurance and a wager. Ex. Policyholders have a strong interest in not dying, hence insurers are prepared to offer life insurance!
- The risk is of a financial nature.
- The risk is reasonably quantifiable, so an insurer is able to assess the risk and set an appropriate premium.
- The probability of the occurrence of the risk is not too high.
- Law of Large Numbers (Video): As the pool of similar (homogeneous) policyholders grows, actual losses become more predictable



Exercise

Your parents have told you that if you marry someone of your choice, they will stop interacting with you. They might even keep you out of their WILL (not give you any share in their property/ wealth)

Can you buy an insurance against this incident?

- What are the different risks involved?
- Are they reasonably quantifiable?
- What is the probability of occurrence?



Insurance Basics

- Information Asymmetry

The insurer can never know as much as the customer. The onus is on the customer to tell everything 'material' to underwriting of her insurance policy.

Information Asymmetry gives rise to two problems

- Adverse Selection: Folks who have reasons to believe that they are at risk will buy more insurance. Since the insurance company is worried that it might not be able to assess the risk properly (the insurance buyer might hide information), it might
 - Keep premium higher to cover higher than actual estimated risk. High price will keep a lot of buyers away from insurance
 - Insurance company might reject a lot of applications

Insurance Basics



- Moral Hazard: The behaviour of the policyholder changes after the policy has been received
 - Insurance policy buyer stops worrying about smoking or drinking more thinking that if anything happens, the hospital expenses will be covered by the health insurance policy.
- **Question:** Will it make sense to sell insurance for losses in stock market investments?
- Investors are careful about losses in the stock market. After buying this policy, they might start speculating in the stock market (Moral Hazard- change of behaviour). The insurance company will get a lot of claims and lose a lot of money.
- Insurance should not cover 'systematic risk' or 'market risk' – if stock markets fall, every investor loses money.



Insurance Basics

Underwriting: Evaluate an applicant's risk level to determine if the insurer should issue a policy and how much premium to charge for it.

Example: Life Insurance

- Financial Underwriting
 - Occupation & Income
 - Calculate 'Human Life Value': the present value of your dependents' future financial needs and includes your future income, expenses, investments, savings, loans and other liabilities.
 - HLV Calculator
- Medical Underwriting
 - Age & Physical Details & Genetic medical history
 - Hobbies & Travel details
 - Occupation: Are you exposed to risks which can impact your health? Ex. harmful fumes in a factory.



Insurance Basics

LIFE INSURANCE: TYPES OF PRODUCTS

- Term Insurance: Only life insurance
 - If the insured person expires, sum assured paid out to the nominee
 - Low premium
- Insurance + Savings
 - Life insurance cover
 - PLUS **Guaranteed** Regular bonus & lump sum when policy matures
 - Higher premium than Pure Term
- Insurance + Investment
 - Life Insurance Cover
 - PLUS investment. Returns vary based on how the investment performs
 - Example: ULIP (Unit Linked Insurance Plan). Operates like (Term Insurance + Mutual Fund)
- Premium varies. Depends on amount that one wants to invest.

Insurance Market in India

Life Insurance

Penetration is the ratio of total insurance premiums to a country's nominal Gross Domestic Product (GDP), expressed as a percentage

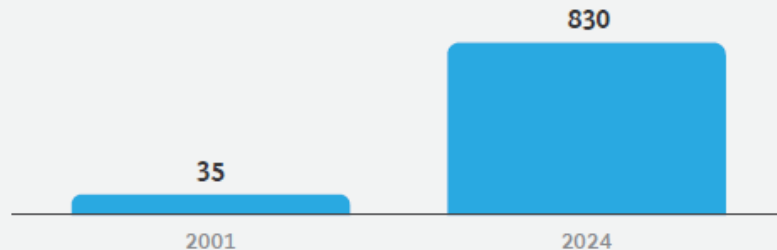
Reference: BCG Report Nov '24



Life

	FY01	FY24
# Players	5	25
Penetration (%) ¹	1.8	2.8
Lives covered (Cr)	5	31 ²

Life Insurance Premiums (₹'000 Cr)

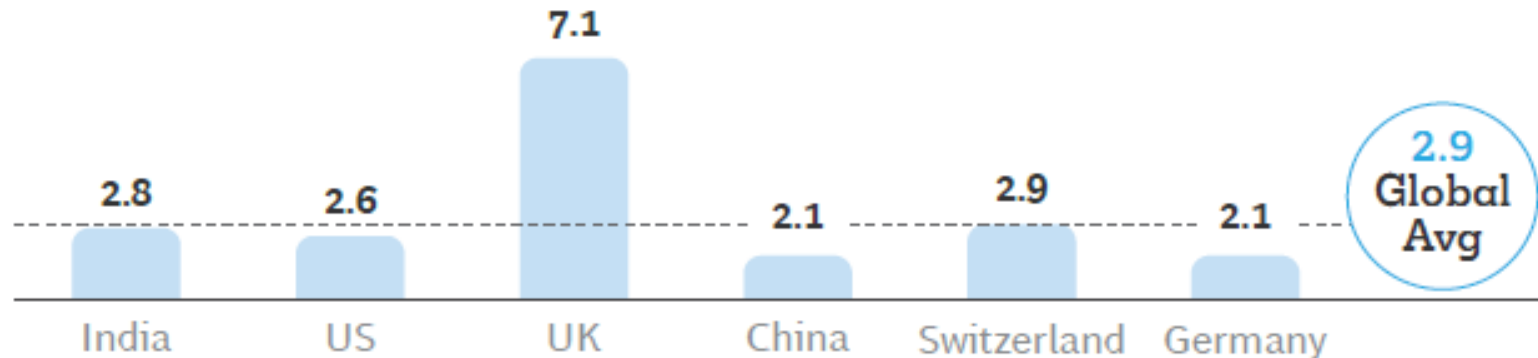




Penetration: Life Insurance

Life Insurance penetration on par with global average

Life Insurance Penetration (% premiums to GDP, 2023)



Non-Life Insurance

Insurance other than Life Insurance. Example: Health, Fire, Marine, Motor, home, factory, shop, travel etc.

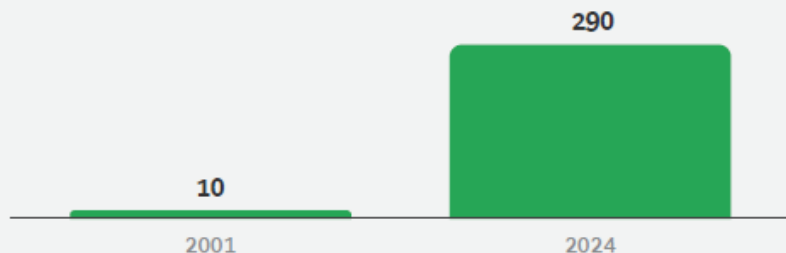
Also called General Insurance in India.



Non-life Insurance

	FY01	FY24
# Players	9	32
Penetration (%) ¹	0.6	1.0
Lives covered (Cr)	4.3	57 ²

Non-life Insurance Premiums (₹'000 Cr)

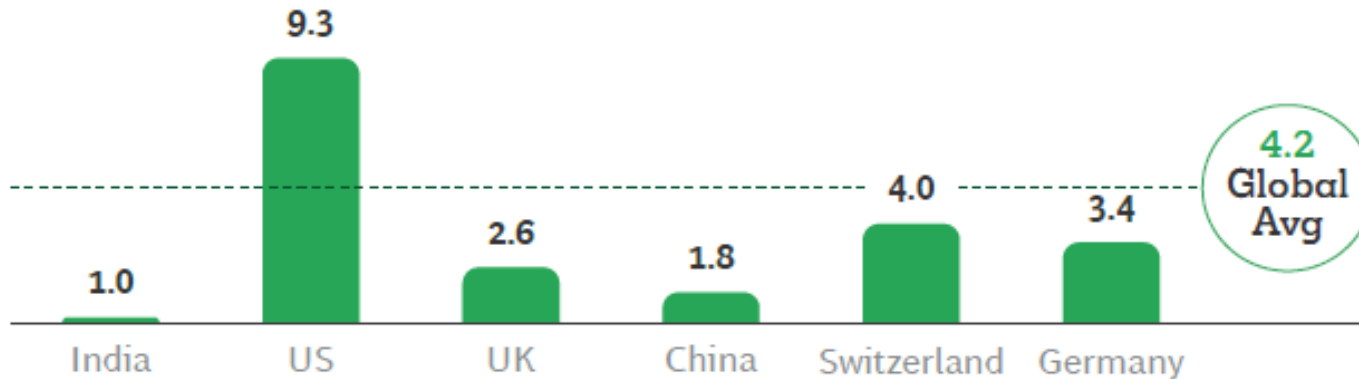




Penetration: Non-Life Insurance

**Need to enhance penetration;
Non-life penetration¹ significantly lower than global average**

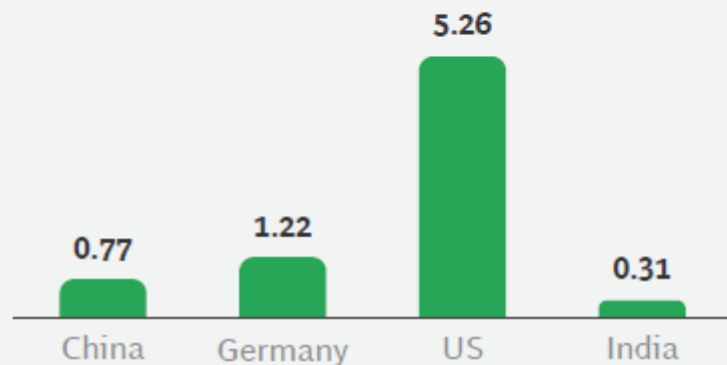
Non-life Insurance Penetration (% premiums to GDP, 2023)



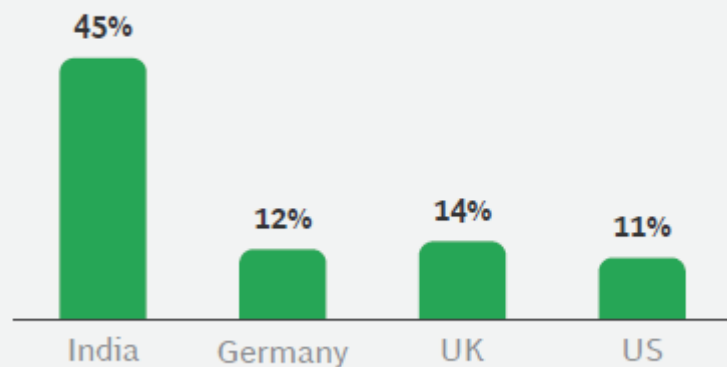
Health Insurance

Health Insurance – Low Penetration¹ and Inadequate Coverage

Health Insurance
Penetration (% premiums
to GDP, 2023)



Out-of-pocket medical
expense as a % of current
health expenditure²



Insurance: Cash Flows,
Investments, Stability



Cash Flows

INSURANCE IS A LONG TERM PRODUCT

- INFLOWS
 - Insurance Premium received
 - Investments products liquidated sold in the market/exchange
 - Claims received from Reinsurance
- OUTFLOWS
 - Claims paid out
 - Expenses: Underwriting, Operational, Distribution
 - Investment products purchased
 - Reinsurance premiums paid
 - Tax paid

Investing Funds Collected



- LONG TERM
 - Government securities, Government guaranteed bonds
 - Equity shares, Preference shares
 - Mutual Funds
 - Derivative Instruments
 - Debentures/ Bonds
 - Fixed Deposits
- SHORT TERM: In addition to the above (of shorter maturities)
 - Govt Securities Including Treasury Bills
 - Repo
 - Commercial Paper
 - Certificate of Deposit



Investments

- To meet guaranteed payouts & estimated claims, large % of investments of Insurance Co. is in Government & Corporate bonds.

OWNERSHIP OF CENTRAL GOVERNMENT BONDS (APPROX.)

- Commercial Banks 38% Why?
 - SLR
- Insurance Companies 26%
 - Explained above + Long term investment needed
- RBI 11%
 - To manage liquidity & interest rates, Banker to the Govt
- Pension Funds 5%
 - To meet guaranteed payouts over the long term
- Mutual Funds 3%
- Others: State Govt, FPIs etc.



Interesting Products

- Credit Life Insurance
 - Insurance for the outstanding loan amount.
 - Decreases as this amount decreases
 - If the borrower expires, the lender gets the insurance payout.
- Credit Default Swaps- CDS
 - Insurance against default on a financial instrument, usually some kind of securitized bond (debt)
 - Holder of debt will buy a CDS from an investment or insurance company, such as AIG, to shift the risk of default to a third party
 - CDS was one of the biggest reason for 2008 crisis
 - CDS AIG 2008 Crisis Video

Reserves & Solvency



TO ENSURE INSURANCE COMPANY FULFILS ITS PAYMENT COMMITMENTS, IT SHOULD MEET THE FOLLOWING FUNDING REQUIREMENTS

- Reserve: An amount, along with future premiums, which is enough to meet the future policy liabilities and future expenses. Co. should have reserves (calculated as below) available.
- $\text{Reserve} = \text{PV of future Policy Benefits} + \text{PV of future policy related expenses} + \text{PV of future Commission/ Incentives} - \text{PV of future Premiums}$
- Solvency Margin: Acts as an additional Cushion (over and above the Reserves) to protect the interests of policyholders
- An insurer must have sufficient capital or surplus assets to meet the statutory minimum solvency margin as specified by the REGULATOR- IRDA (Insurance Regulatory & Development Authority)
- Somewhat similar in concept to CRR, SLR in banking.

Pension



Pension Basics

- A pension plan is an asset pool that accumulates over an individual's working years and is paid out during the non-working years
- A growing market because:
 - People in old age cannot depend on kids or their larger family
 - People expecting to live longer & retiring earlier
- Pension Fund Managers in India
 - SBI Pension Funds
 - LIC Pension Fund
 - UTI Retirement Solutions
 - HDFC Pension Management
 - ICICI Prudential Pension Fund Management
 - Etc.



Pension: Types

- Defined Benefit Pension Plan: The plan sponsor (employer, government) promises the employees a specific benefit when they retire.
- Defined Contribution Pension Plan: It specifies what will be contributed to the pension fund. Later payouts will depend on what the investment returns are.
- Pension can be provided by the government, any other employer or a person can buy for himself/herself.
- Regulator in India: PFRDA Pension Fund Regulatory & Development Authority
- Investment should yield returns to take care of future payouts
- Cash flows from investments done by Pension Manager should match the cash flows required for **annuity payouts**.

Annuity



- An annuity plan provides guaranteed regular payments during retirement for the rest of your life in exchange for a lump sum payment or a series of instalments.
- Accumulation Phase: When you pay
- Distribution Phase: When you receive income
- Often a person who saves using insurance or pension fund can buy an annuity when the policy matures.
- Annuity can stop completely when the person expires or can continue for the spouse or the remaining amount be given to descendants.
- Sold by life insurance companies and pension companies

National Pension Scheme



- NPS- Introduced by the Central Government to help individuals have income in the form of pension to take care of their retirement needs.
- All Pension Funds offer their especially designed products through NPS
- PFRDA regulates and administers NPS
- Costs kept low by PFRDA. Any individual can buy NPS.
- NPS is a market-linked **defined contribution** scheme
- All asset classes- Equity, Gov bonds, corporate bonds, Alternative Investments available. Buyer can choose how she wants to invest her funds.
- Tax benefits given on investment in NPS to encourage more buyers
- About NPS Returns from NPS



Investments

- Majority of investment of Pension Funds is in Bonds.
- HDFC Pension, Annual Report, FY 2025-25
 - 8.32% GOI, August 02, 2032
 - 9.20% GOI, September 30, 2030
 - 7.48% Andhra Pradesh SDL Mat 03-Sep-2034
 - 6.64% GOI, June 16, 2035
 - 6.75% Gujarat SDL, October 13, 2029
 - 8.15% Power Grid, March 09, 2030 NCD
 - 8.27% National Highways Authority of India ,March 28, 2029 NCD
 - 7.42% SIDBI NCD Mat 12-Mar-2029



Investments

- Internationally, Pension funds have large AUMs.

Government Pension Investment Fund (GPIF), Japan \$ 1.8 Trillion

Government Pension Fund Global, Norway \$ 1.8-2 trillion

Canada Pension Plan Investment Board, Canada \$ 1.5 trillion

California Public Employees' Retirement System, USA \$ 0.5 trillion

- Largest Insurance Companies

Allianz, Germany \$ 2.6 trillion

AXA, France \$ 1.1 trillion

Prudential Financial, USA \$1.5 trillion

LIC of India \$700 billion

Summary



- Insurance & Pension – both are long term products. Hence highly regulated to ensure the companies estimate risks properly and have enough capital to fulfil payment commitments
- To meet guaranteed payouts & estimate claims, large % of investments of Insurance Co. and Pension Funds is in Government & Corporate bonds.
- These firms are an important source of borrowings for the government and private sector & play an important role in Financial Mkts.
- When the Bond Markets Made Trump Rethink



Thank You!